

The Demographic Health Index Methodology

Version 2.0 · June 2026

Demoria Research · demoriaresearch.com/methodology

Cite as: Demoria Research (2026). The Demographic Health Index: Methodology (v2.0). demoriaresearch.com/methodology. DOI: pending.

Licence: free to reuse with attribution for journalism, research and education; commercial licensing by enquiry. demoriaresearch.com/licence

What the DHI measures

It measures one thing: the arithmetic of replacement. Are enough children being born to replace the people already here? Can the age structure carry its dependents? And is stability held by a country's own momentum, or rented from the rest of the world through migration?

The DHI is not a measure of wealth, happiness, or the worth of a society. A rich country can score badly; a poor one can score well. The index is deliberately blind to income — it asks only whether a population can sustain itself.

The usual shorthand for that question is the headline fertility rate. It flatters and it misleads. A total fertility rate says nothing about the age structure beneath it, nothing about how long a country has already been below replacement, and nothing about the difference between a society carried by its own young cohorts and one carried by a constant inflow of migrants.

Two countries with an identical TFR can be on opposite trajectories. Version 2.0 was rebuilt from the ground up to pull those forces apart — into four weighted pillars and fifteen indicators.

Every figure rests on the UN World Population Prospects 2024 revision. Where a national statistical office reports materially different fertility, that value is applied as an explicit override — and every country-year is stamped WPP or NSO, so provenance is never hidden.

02 • WHY IT MATTERS

So what?

Roughly two-thirds of humanity now lives in a country with below-replacement fertility. The defining demographic story of this century is not growth — it is the depth, the speed and the reversibility of decline.

These are slow variables with hard consequences. A working-age base that shrinks under a growing retired population reprices pensions, healthcare, housing, defence and the tax take all at once — and labour-force contraction compounds on itself. By the time the strain is obvious in the headline numbers, the cohorts who would have eased it were never born.

The index separates a country that is genuinely self-sustaining from one that is merely being held up. Two props can flatter a headline. Momentum — a young age structure keeps births high for a generation even after fertility falls below replacement; it buys time, it does not last. Migration — a contingent prop that depends on the rest of the world continuing to supply people; it can hold a population flat indefinitely, but it is borrowed, not owned.

So South Korea and Niger sit near opposite ends of the index, and the reasons compound rather than cancel. Several Gulf states post stable working-age numbers built almost entirely on a guest-worker inflow that is neither settling nor reproducing locally. A handful of micro-jurisdictions look buoyant only because tax and residency rules import the very people the headline counts. The DHI is built to name these mechanisms — not to be flattered by them.

It is not a forecast of national success, a ranking of desirability, or a verdict on migration, family policy or any group of people. It is a structural measurement of one variable — replacement — and the uncertainty around it is shown, not hidden: the Low and High variants bracket every projection, and every input traces to a public source.

03 · HOW IT IS BUILT

The method

The DHI is a weighted geometric mean of four pillars, each built from a set of indicators, computed for every country and year and rescaled to a readable 0–100.

The pillars are combined geometrically, not averaged. An average lets a strong pillar paper over a collapsed one — a country could mask failed fertility behind heavy migration. A geometric mean punishes imbalance: if any pillar falls toward zero, the whole score follows it down. For a sustainability measure, the weakest link should govern.

UN World Population Prospects 2024 is the global baseline. Where a national statistical office reports a materially different total fertility rate, that value flows in through an explicit cascade — into the net reproduction rate, the birth stream and natural change — so the override stays consistent across every indicator it touches.

Scores are observed annually from 1965 to 2025, then projected to 2100 under the UN's Low, Median and High variants. The headline figure uses the Median; the World view exposes all three at four waypoints — 2035, 2050, 2075 and 2100.

Because fertility is scored on a curve that also penalises births far above replacement, the High-fertility variant does not always yield the highest score. For an already-high-fertility country it can sit below the Median path. The variant selector shows a scenario, not a simple best-to-worst ladder.

Two countries can share a score for opposite reasons. Beneath the number, each is assigned one mechanism-based class — the number tells you how healthy, the typology tells you why.

Two further sets of indicators are surfaced for every country — not as part of the score, but next to it. Fertility Quality covers adolescent fertility, infant mortality and the share of births to women 35+. Human Capital covers mean years of schooling 20–39, the share of adults with upper-secondary education and the gender gap in schooling. They explain the human and developmental context behind a demographic score — a country can sit in the Collapsing band with a world-class human-capital base at the same time, and the two together are more honest than either alone.

04 · THE AUTHOR

About the author

Harry studied economics and demography at the University of Bath (BSc), Bocconi University (MSc) and Tsinghua University (MSc exchange semester). His personal interest is in fertility collapse and demographic ageing — what he considers the most significant and

furthest-reaching macro-trend in the world today.

Previous work has spanned the start-up space, financial services, and the commodities industry — most recently at Onyx Capital Group and ArcelorMittal — where the demand-side consequences of demographic change are routinely underweighted by the markets and institutions they will eventually reshape.

He was born in the UK, but has studied in Italy and China.

Data: UN World Population Prospects 2024; national statistical offices;
Demoria Research estimations. Live data: demoriaresearch.com/births ·
profiles: demoriaresearch.com/country